Reproducible Report on COVID19 Data

Report

2026-10-14

Introduction

The COVID-19 pandemic affected every country in the world. To monitor its course, between January 2020 and March 2023, the Center for Systems Science and Engineering (hereafter abbreviated as CSSE) at Johns Hopkins University maintained a dataset compiled from data retrieved from national health authorities, downloadable from a single GitHub repository [1], [2]. This brief report presents the main findings from its global time series of cumulative confirmed cases and deaths by country and answers three research questions: (1) what were the long-term trends of cumulative cases, deaths, and, above all, the mortality rate (total deaths to total cases, i.e., the case fatality rate) for the world as a whole (aggregated to monthly frequency to employ time-series models), (2) were cases, deaths, and the mortality rate distributed uniformly across countries, and (3) which (preferably, automatically selected) ARIMA model describes the monthly world series and what does it predict for the subsequent 12 months. Each question is complemented with a formal statistical hypothesis (or model for the Auto-ARIMA) tested at the standard 5% significance level ($\alpha = 0.05$):

- **H1**. Monthly cumulative cases, deaths, and the mortality rate have a statistically significant linear trend, i.e., $\beta_1 \neq 0$ in $y_t = \beta_0 + \beta_1 t$, at both world (total of all countries) and individual country level.
- **H2**. Across countries, cases and deaths are distributed uniformly, and the mortality rate is uniform (i.e., deaths are distributed proportionally to cases), under a built-in Pearson's χ^2 goodness-of-fit test.
- **H3**. The monthly world time series are adequately described and afterward predicted (for at least the subsequent 12 months) by an $ARIMA(p, d, q)(P, D, Q)_{12}$ process, selected automatically by the Hyndman-Khandakar algorithm [3], i.e., the model residuals are white noise under the Ljung-Box test.

To compile this report, R packages `tidyverse`, `lubridate`, `rconvertu` (the author's package converting between country names and ISO 3166-1 alpha-3, alpha-2, and numeric codes using regular expressions), `ggplot2`, and `forecast` were employed, with all R code embedded into the R Markdown document to ensure its reproducibility. The CSV datasets, downloaded from the CSSE GitHub repository, are accessed in the folder `./DATA/` as `time_series_covid19_confirmed_global.csv` and `time_series_covid19_deaths_global.csv` (the detailed US files are ignored). The report is subdivided into five parts: data description and processing, time trends, cross-country structure, (ARIMA) time-series model with a 12-month forecast, and potential bias.

Data Description and Processing

The raw (unprocessed) dataset consists of two CSV files (confirmed cumulative cases and deaths), each with 289 rows and 1,147 columns, i.e., 201 countries and 91 provinces or territories thereof, with one row per territorial unit and one column per day between January 22, 2020 and March 09, 2023 (i.e., the dataset is in a wide format). Given that the data comprise 404 missing values (all in `Province/State`), daily cumulative counts (which are too detailed for the intended time-series analysis), unused columns (`Lat`, `Long`), non-ISO 3166-1 country names (e.g., "US", "Korea, South", or "Taiwan*"), and minor value inconsistencies (e.g.,"Congo (Brazzaville)" and "West Bank and Gaza" must be recoded for the ISO 3166-1 conversion), the raw dataset required several preprocessing steps – employing `tidyverse`'s `pivot_longer()`, `full_join()`, `select()`, `mutate()`, `group_by()`, `summarize()`, and `slice_max()` functions, alongside `rconvertu`'s `convertu()` for ISO 3166-1 alpha-3 codes (`ISO3`) and English names (`NAME_EN`) – to obtain a modified panel tibble `COVID` in a long format (month-end cumulative values) and the world totals `WORLD` (consult the R code field below).

```
COVID <- read_csv('./DATA/time_series_covid19_confirmed_global.csv')          %>%
         pivot_longer(-(1:4), names_to = "DATE", values_to = "CASES")        %>%
         full_join(
         read_csv('./DATA/time_series_covid19_deaths_global.csv')            %>%
         pivot_longer(-(1:4), names_to = "DATE", values_to = "DEATHS")       %>%
         select(-c(Lat, Long)),
         by = c("Province/State", "Country/Region", "DATE"))                 %>%
         select(  COUNTRY   = `Country/Region`, DATE, CASES, DEATHS)         %>%
         mutate(  COUNTRY   = recode(COUNTRY,
                              "Congo (Brazzaville)" = "Republic of the Congo",
                              "West Bank and Gaza"  = "Palestine"),
                  DATE      = mdy(DATE),
                  MONTH     = floor_date(DATE, "month"))                      %>%
         group_by(COUNTRY, MONTH, DATE)                                      %>%
                  summarize(CASES     = sum(CASES,  na.rm = TRUE),
                            DEATHS    = sum(DEATHS, na.rm = TRUE),
                            .groups   = "drop")                              %>%
         group_by(COUNTRY, MONTH)                                            %>%
                  slice_max(DATE, n = 1, with_ties = FALSE)                  %>%
         group_by(COUNTRY)                                                   %>%
         mutate(  ISO3      = convertu(to = "iso3",    text = first(COUNTRY)),
                  NAME_EN   = convertu(to = "name_en", text = first(COUNTRY)),
                  ISO3      = factor(if_else(ISO3 == COUNTRY, NA, ISO3)),
                  MORTALITY = if_else(CASES > 0, DEATHS / CASES, NA))        %>%
         ungroup()                                                           %>%
         filter(  !is.na(ISO3))                                              %>%
         select(  ISO3, NAME_EN, MONTH, CASES, DEATHS, MORTALITY)            %>%
         arrange( ISO3, MONTH)

WORLD <- COVID                                                               %>%
         group_by(MONTH)                                                     %>%
                  summarize(CASES     = sum(CASES,  na.rm = TRUE),
                            DEATHS    = sum(DEATHS, na.rm = TRUE),
                            .groups   = "drop")                              %>%
         mutate(  MORTALITY = DEATHS / CASES)
```

The obtained panel tibble `COVID` consists of 7,683 rows (197 countries × 39 months) and 6 columns, with 508 missing values (`MORTALITY` is undefined without confirmed cases); 4 units without an ISO 3166-1 code (Diamond Princess, MS Zaandam, Summer Olympics 2020, Winter Olympics 2022) were dropped. Between January 2020 and March 2023, **676,568,028 confirmed cases** and **6,881,787 deaths** were reported, i.e., a **mortality rate of 1.02%**, with the most cases and deaths in **United States of America (the)**, and the highest mortality rate (among countries with over 1,000 cases) in **Yemen**. The `summary()` output is below.

```
summary(COVID)
```

```
##       ISO3        NAME_EN              MONTH               CASES
##  AFG    :  39   Length:7683        Min.   :2020-01-01   Min.   :        0
##  ALB    :  39   Class :character   1st Qu.:2020-10-01   1st Qu.:     5079
##  DZA    :  39   Mode  :character   Median :2021-08-01   Median :    61659
##  AND    :  39                      Mean   :2021-08-01   Mean   :  1464101
##  AGO    :  39                      3rd Qu.:2022-06-01   3rd Qu.:   537822
##  ATA    :  39                      Max.   :2023-03-01   Max.   :103802702
##  (Other):7449
##      DEATHS          MORTALITY
```

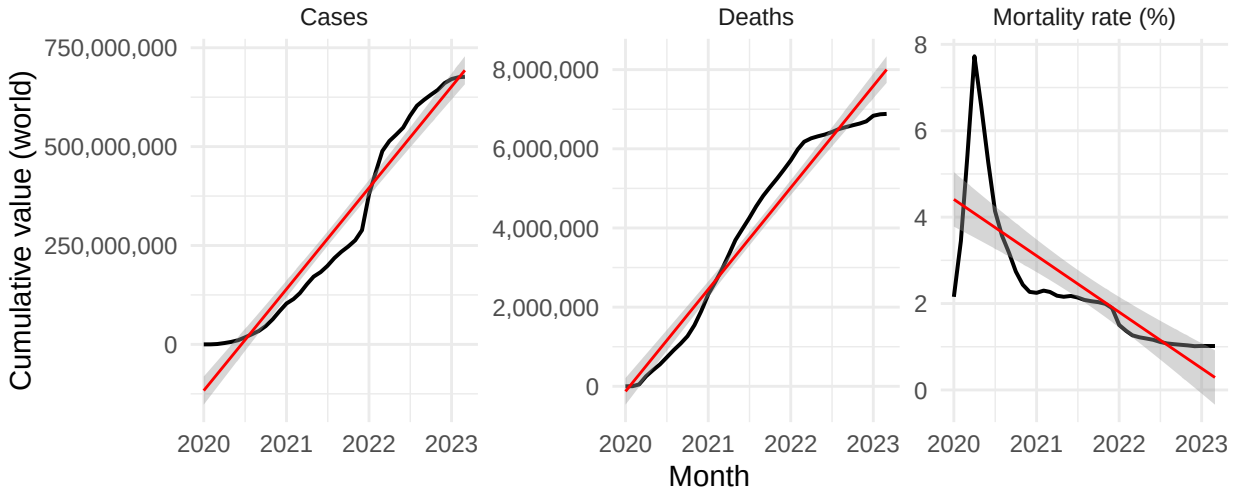
```
##  Min.   :      0   Min.   :0.00000
##  1st Qu.:     61   1st Qu.:0.00709
##  Median :    846   Median :0.01429
##  Mean   :  20007   Mean   :0.02956
##  3rd Qu.:   7694   3rd Qu.:0.02437
##  Max.   :1123836   Max.   :6.00000
##                    NA's   :508
```

Analysis of Time Trends

Aggregated monthly data show an increasing (cumulative) number of cases and deaths worldwide, with the steepest increase in cases during the so-called Omicron wave in early 2022 (see Figure 1). The mortality rate, on the contrary, peaked at 7.72% in April 2020, when testing was scarce and limited to severe cases only, and then declined steadily towards 1.02% in March 2023 due to advances in mitigation (among other, vaccine roll-out), reducing the disease to milder variants. Both cumulative curves also flattened in late 2022 and early 2023, when many countries scaled down reporting (the CSSE's last dataset update was on March 10, 2023).

```
ggplot(WORLD                                                                %>%
       transmute(MONTH, Cases = CASES, Deaths = DEATHS,
                 `Mortality rate (%)` = 100 * MORTALITY)                    %>%
       pivot_longer(-MONTH, names_to = "SERIES", values_to = "VALUE")       %>%
       mutate(SERIES = factor(SERIES, levels = c("Cases", "Deaths",
                                                 "Mortality rate (%)"))),
       aes(x = MONTH, y = VALUE))                                             +
  geom_line(linewidth = 0.7)                                                  +
  geom_smooth(method = "lm", formula = y ~ x, linewidth = 0.5, color = "red") +
  facet_wrap(~SERIES, scales = "free_y", axes = "all_x",
                                         axis.labels = "all_x")               +
  scale_y_continuous(labels = scales::comma)                                  +
  labs(
    title = paste0("Figure 1. Cases, deaths, and mortality rate, ",
                   format(min(WORLD$MONTH), "%b %Y"), "-",
                   format(max(WORLD$MONTH), "%b %Y")),
    x = "Month",
    y = "Cumulative value (world)"
  )                                                                           +
  theme_minimal()
```

Figure 1. Cases, deaths, and mortality rate, Jan 2020−Mar 2023



The formal statistical test of **H1** is tied to coefficients β_1 from three linear regression models `lm(VALUE ~ MONTH SEQUENCE)`, i.e., for the cumulative cases, deaths, and the mortality rate of the world as a whole, with the `MONTH` encoded with the help of `seq_along()` into a time-series sequence. Since the CSSE data contain no regional (continental) aggregates, the same model was also estimated for the mortality rate of each of the 195 countries with at least 12 months of data: the slope is significantly negative for 113, significantly positive for 38, and insignificant for 44 countries, i.e., the declining world trend is shared by a majority (58%) of countries. Table 1 comprises the slopes, their standard errors, the two-sided p-values, and the R^2 of the fit: all three coefficients are non-zero at the 5% significance level ($\alpha = 0.05$), with the mortality rate declining on average by 0.109 percentage points per month. However, the high R^2 values of cumulative series are typical of integrated (non-stationary) processes, i.e., potential unit roots to be addressed by an (S)ARIMA model.

```
WORLD                                                                    %>%
transmute(MONTH, Cases = CASES, Deaths = DEATHS,
          `Mortality rate (%)` = 100 * MORTALITY)                        %>%
pivot_longer(-MONTH, names_to = "SERIES", values_to = "VALUE")           %>%
mutate(  SERIES = factor(SERIES, levels = c("Cases", "Deaths",
                                            "Mortality rate (%)")))      %>%
 arrange(SERIES, MONTH)                                                  %>%
group_by(SERIES)                                                         %>%
         summarize(MODEL = list(summary(lm(VALUE ~ seq_along(MONTH)))))  %>%
transmute(
         Series  = SERIES,
         Slope   = sapply(MODEL, \(m) m$coefficients[2, 1]),
         SE      = sapply(MODEL, \(m) m$coefficients[2, 2]),
         P       = sapply(MODEL, \(m) m$coefficients[2, 4]),
         R2      = sapply(MODEL, \(m) m$r.squared)
)                                                                        %>%
mutate(P = scales::pvalue(P, accuracy = 0.001))                          %>%
knitr::kable(digits  = 3, format.args = list(big.mark = ",",
                                             scientific = FALSE),
             caption = "Linear trends in world cases, deaths, and mortality")
```

Table 1: Linear trends in world cases, deaths, and mortality

Series	Slope	SE	P	R2
Cases	21,337,363.468	793,201.703	<0.001	0.951
Deaths	214,167.931	7,521.517	<0.001	0.956
Mortality rate (%)	-0.109	0.014	<0.001	0.616

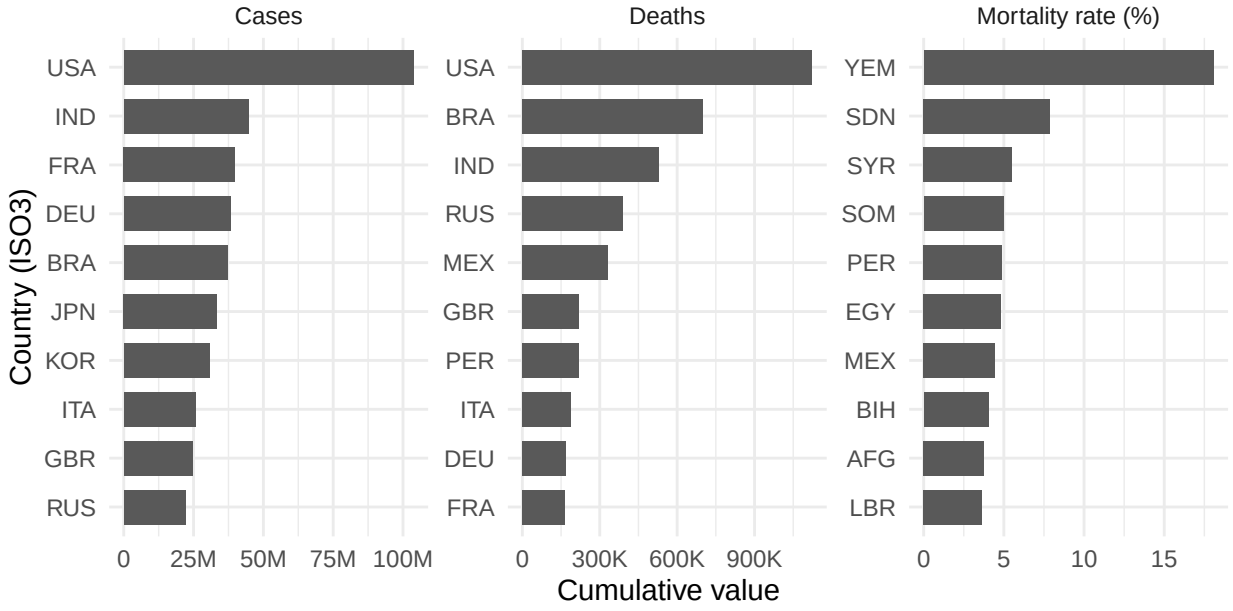
Analysis of Cross-Country Structure

Figure 2 demonstrates the distribution of cases, deaths, and the mortality rate by country (labeled with ISO 3166-1 alpha-3 codes) as of March 2023. Namely, the three countries with the most cases are **United States of America (the), India, France**, with the most deaths **United States of America (the), Brazil, India**, and with the highest mortality rate (among countries with over 1,000 cases, which excludes a handful of micro-states with implausible ratios) **Yemen, Sudan (the), Syrian Arab Republic (the)**. Whereas the first two rankings are dominated by populous countries, the third one comprises mostly low-income or conflict-affected countries, i.e., strongly indicating a non-uniform distribution of the examined COVID data.

```
ggplot(LATEST                                                            %>%
       transmute(ISO3, Cases = CASES, Deaths = DEATHS,
                 `Mortality rate (%)` = if_else(CASES >= 1000,
                                                100 * MORTALITY, NA))     %>%
       pivot_longer(-ISO3,  names_to  = "SERIES", values_to = "VALUE")   %>%
```

```r
    filter(!is.na(VALUE))                                                    %>%
    mutate(SERIES = factor(SERIES, levels = c("Cases", "Deaths",
                                              "Mortality rate (%)")))        %>%
    group_by(SERIES)                                                         %>%
             slice_max(order_by = VALUE, n = 10, with_ties = FALSE)          %>%
    ungroup(),
    aes(x = reorder(paste(ISO3, SERIES, sep = "|"), VALUE), y = VALUE))      +
geom_col(width = 0.7)                                                        +
coord_flip()                                                                 +
facet_wrap(~ SERIES, scales = "free", axes        = "all_x",
                                      axis.labels = "all_x")                 +
scale_x_discrete(labels = \(x) str_remove(x, "\\|.*$"))                      +
scale_y_continuous(labels = scales::label_number(
                            scale_cut = scales::cut_short_scale()))          +
labs(
  title = paste0("Figure 2. Top ten countries (ISO 3166-1 codes), ",
                 format(max(COVID$MONTH), "%b %Y")),
  x = "Country (ISO3)",
  y = "Cumulative value"
)                                                                            +
theme_minimal()
```

Figure 2. Top ten countries (ISO 3166−1 codes), Mar 2023



The formal statistical test of **H2** is based on Pearson's chi-square goodness-of-fit test `chisq.test()` applied to three distributions across countries as of March 2023, i.e., for the world as a whole (regional aggregates being absent from the data). For cases and deaths, the null hypothesis assumes a uniform distribution across the k countries ($H_0 : p_1 = p_2 = \cdots = p_k = 1/k$); for the mortality rate, it assumes deaths distributed proportionally to cases ($H_0 : p_i = \text{CASES}_i / \sum_{j=1}^{k} \text{CASES}_j$), i.e., an identical mortality rate in every country, with a p-value precision of 0.001. Table 2 comprises the χ^2 statistics, degrees of freedom, and upper-tail p-values. H_0 is rejected in all cases at the 5% significance level ($\alpha = 0.05$), confirming a statistically significant non-uniform distribution across countries, which may reflect variations in population size, age structure, testing capacity, healthcare quality, reporting standards, etc. – i.e., potential relationships that require further investigation.

```
LATEST                                                                  %>%
summarize(
         Cases            = list(chisq.test(CASES)),
         Deaths           = list(chisq.test(DEATHS)),
        `Mortality rate`  = list(chisq.test(DEATHS, p = CASES / sum(CASES)))
)                                                                       %>%
pivot_longer(everything(), names_to = "SERIES", values_to = "TEST")     %>%
transmute(
         Series  = SERIES,
         Chi2    = sapply(TEST, \(m) unname(m$statistic)),
         DF      = sapply(TEST, \(m) unname(m$parameter)),
         P       = sapply(TEST, \(m) m$p.value)
)                                                                       %>%
mutate(P = scales::pvalue(P, accuracy = 0.001))                         %>%
knitr::kable(digits  = 3, format.args = list(big.mark = ",",
                                             scientific = FALSE),
             caption = "Pearson chi-square tests of uniform distribution")
```

Table 2: Pearson chi-square tests of uniform distribution

Series	Chi2	DF	P
Cases	5,953,407,513	196	<0.001
Deaths	70,182,456	196	<0.001
Mortality rate	4,324,734	196	<0.001

Time-Series Model and Forecast

In order to describe the dynamics of the three monthly world series and to forecast their values for the 12 months following March 2023 – which may provide valuable insights for public health authorities in anticipating the burden of the disease – the following seasonal autoregressive integrated moving average model, $ARIMA(p,d,q)(P,D,Q)_{12}$, is employed (formalized in its standard backshift-operator-based form):

$$\Phi_P(B^{12})\,\phi_p(B)\,(1-B)^d\,(1-B^{12})^D\,y_t = c + \Theta_Q(B^{12})\,\theta_q(B)\,\varepsilon_t, \qquad \varepsilon_t \sim WN(0,\sigma^2),$$

where B is the backshift operator, ϕ_p, θ_q, Φ_P, and Θ_Q are the (seasonal) autoregressive and moving average polynomials, and d and D are the orders of (seasonal) differencing. The orders are selected automatically by the Hyndman-Khandakar algorithm implemented in `forecast::auto.arima()` [3], [4], which determines d and D with unit-root tests and then performs a stepwise search over p, q, P, and Q to minimize the corrected Akaike information criterion (AIC). The model is estimated on the tibble `WORLD` for cases, deaths, and the mortality rate (in percent) separately (see Table 3, Figure 3, and the R code below for additional details).

```
models    <- WORLD                                                      %>%
             transmute(Cases = CASES, Deaths = DEATHS,
                       `Mortality rate (%)` = 100 * MORTALITY)          %>%
             map(\(y) auto.arima(ts(y, frequency = 12,
                                    start = c(year( min(WORLD$MONTH)),
                                              month(min(WORLD$MONTH))))))
forecasts <- map(models, forecast, h = 12)
```

Following the model, the formal statistical test of **H3** is the Ljung-Box test of the residuals, $H_0 : \rho_1 = \cdots = \rho_h = 0$ (white noise) against $H_1 : \exists_{\rho_j}\ \rho_j \neq 0$, with h and the degrees of freedom chosen as in `forecast::checkresiduals()`. Table 3 comprises the selected specification of each model, its AIC, and the Ljung-Box statistic Q, degrees of freedom, and p-value. `auto.arima()` selected ARIMA(1,1,0) with drift

for cases, ARIMA(0,2,0) for deaths, and ARIMA(3,1,0) for the mortality rate, i.e., all three series required differencing, as anticipated in the analysis of time trends, while no seasonal components were selected. H_0 is not rejected at the 5% significance level ($\alpha = 0.05$), i.e., the residuals of all three models are statistically indistinguishable from white noise, supporting **H3**, i.e. making the Auto-ARIMA to be a good model choice.

```
DIAG                                                                    %>%
mutate(P = scales::pvalue(P, accuracy = 0.001))                         %>%
knitr::kable(digits  = 3, format.args = list(big.mark = ",",
                                             scientific = FALSE),
             caption = "Selected ARIMA models and Ljung-Box tests")
```

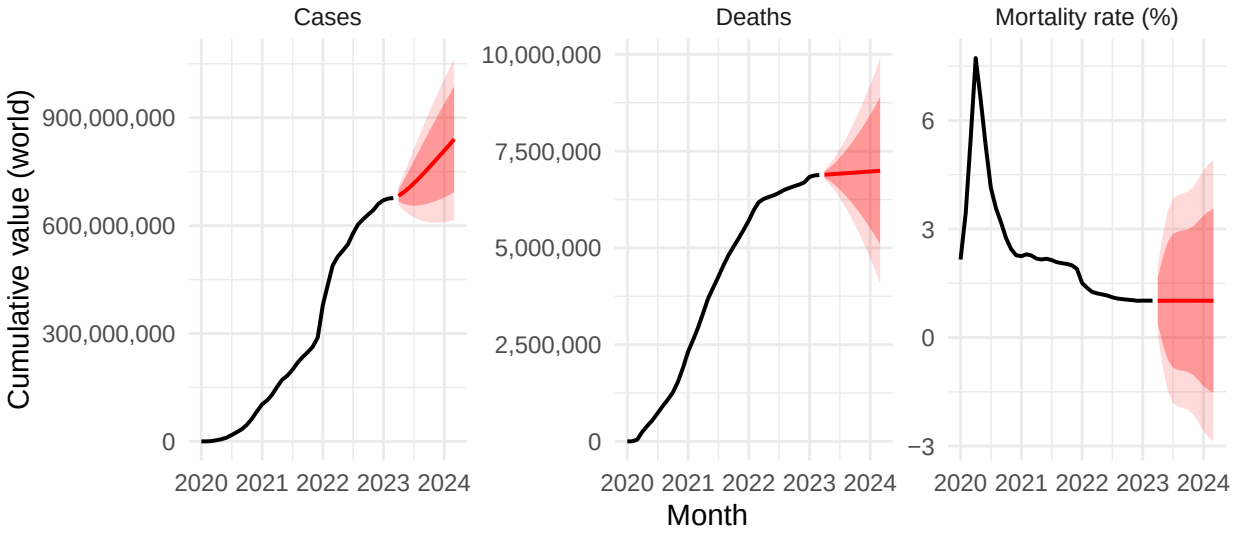
Table 3: Selected ARIMA models and Ljung-Box tests

Series	Model	AIC	Q	DF	P
Cases	ARIMA(1,1,0) with drift	1,356.856	4.520	6	0.607
Deaths	ARIMA(0,2,0)	919.142	5.565	8	0.696
Mortality rate (%)	ARIMA(3,1,0)	61.533	6.618	5	0.251

Figure 3 depicts the 12-month forecasts (point forecasts with 80% and 95% prediction intervals), which extrapolate the cumulative time series: by March 2024, the point forecasts amount to ca. **840,347,772 cases** and **6,991,227 deaths**, with the **mortality rate at 1.02%** (versus 1.02% in March 2023), while the implied ratio of the forecasted deaths to the forecasted cases declines to ca. 0.83%, since the forecasted cases grow faster – i.e., the three univariate models are not mutually consistent. Given only 39 monthly observations, the prediction intervals widen rapidly with the horizon (for the mortality rate, the lower bounds even become negative, since the Gaussian intervals ignore the non-negativity of the ratio), and the forecasts should be interpreted as a continuation of the most recent dynamics rather than a serious epidemiological projection.

```
ggplot(WORLD                                                            %>%
       transmute(MONTH, Cases = CASES, Deaths = DEATHS,
                 `Mortality rate (%)` = 100 * MORTALITY)                %>%
       pivot_longer(-MONTH, names_to = "SERIES", values_to = "VALUE")   %>%
       mutate(SERIES = factor(SERIES, levels = names(models))),
       aes(x = MONTH, y = VALUE))                                         +
  geom_ribbon(data = FORECAST, aes(ymin = LOW95, ymax = HIGH95),
              fill = "red", alpha = 0.15)                               +
  geom_ribbon(data = FORECAST, aes(ymin = LOW80, ymax = HIGH80),
              fill = "red", alpha = 0.30)                               +
  geom_line(data = FORECAST,  linewidth = 0.7,  color = "red")          +
  geom_line(linewidth = 0.7)                                            +
  facet_wrap(~SERIES, scales = "free_y", axes = "all_x",
                                         axis.labels = "all_x")         +
  scale_y_continuous(labels  = scales::comma)                           +
  labs(
    title = paste0("Figure 3. ARIMA forecasts, ",
                   format(min(FORECAST$MONTH), "%b %Y"), "-",
                   format(max(FORECAST$MONTH), "%b %Y")),
    x = "Month",
    y = "Cumulative value (world)"
  )                                                                     +
  theme_minimal()
```

Figure 3. ARIMA forecasts, Apr 2023−Mar 2024



Discussion of Potential Bias

The findings in the report, despite the use of rigorous methodology, are subject to several potential biases, the first group of which is data-related. On the one hand, both the numerator and the denominator of the mortality rate are measured with error: COVID-19 deaths may have included deaths "with" rather than "from" the disease (e.g., patients with severe comorbidities or other diseases, since countries differed in whether a positive test alone sufficed for the classification), while confirmed casescomprised false positives (imperfect specificity of PCR and rapid antigen tests) and, far more importantly, omitted a substantial share of undetected cases (asymptomatic or mild infections, limited testing capacity, and individual home testing from 2022 onward). On the other hand, monthly aggregation discarded the within-month variability in data.

The second group of potential biases is method-related, given that the three hypotheses were tested with basic (textbook) procedures. First, the linear trends in Table 1 were estimated on cumulative (integrated) series, which violate the Gauss-Markov assumptions and inflate both the R^2 and the significance of the slopes – i.e., the trends are descriptive rather than inferential. Second, Pearson's χ^2 tests in Table 2 are based on counts in the hundreds of millions, so that even thr smallest deviations are statistically significant, i.e., the rejection of **H2** is practically guaranteed (moreover, the expected counts of a few micro-states fall below five). Third, the ARIMA models were estimated on only 39 monthly observations with structural breaks (new disease variants, vaccination, and/or changes in mitigation policies), and the mortality rate was modeled directly as a ratio rather than derived from its components. Hence, although these limitations do not necessarily invalidate the obtained findings, they merit further investigation (beyond the scope of this short report).

Conclusion

This brief report examined 676,568,028 confirmed COVID-19 cases and 6,881,787 deaths recorded by the Johns Hopkins University CSSE in 197 countries between January 2020 and March 2023 (a world mortality rate of 1.02%) and answered the three research questions with formal statistical hypotheses tested at the 5% significance level ($\alpha = 0.05$). First, in terms of long-term trends (1), the monthly series exhibited significant linear trends – increasing for cumulative cases and deaths and declining for the mortality rate, which peaked at 7.72% in April 2020 – leading to acceptance of **H1** (see Table 1), with the declining trend shared by 113 of the 195 countries. Second, in terms of the cross-country structure (2), cases and deaths were concentrated in a few populous countries, while the highest mortality rates were observed in countries with limited testing capacity (see Figure 2), and Pearson's χ^2 tests rejected both the uniform distribution of cases and deaths and the proportionality of deaths to cases (see Table 2), i.e., rejecting **H2**. Third, in terms of the time-series model (3), `auto.arima()` selected ARIMA(1,1,0) with drift, ARIMA(0,2,0), and ARIMA(3,1,0) for cases,

deaths, and the mortality rate, respectively, with the Ljung-Box tests supporting **H3** (see Table 3), and the 12-month forecasts extrapolated the cumulative series, with the mortality rate expected at ca. 1.02% by March 2024 (see Figure 3). To sum up, in light of the potential biases – above all, the measurement error in both deaths (which may have included other diseases) and cases (false positives and, predominantly, undetected infections) – the findings should be interpreted with caution and subject to further investigation.

References

[1] Dong, Ensheng, Hongru Du, and Lauren Gardner. 2020. "An Interactive Web-Based Dashboard to Track COVID-19 in Real Time." *The Lancet Infectious Diseases* 20 (5): 533–34. https://doi.org/10.1016/S1473-3099(20)30120-1.

[2] Johns Hopkins University Center for Systems Science and Engineering (JHU CSSE). 2023. "COVID-19 Data Repository by the Center for Systems Science and Engineering (CSSE) at Johns Hopkins University." Dataset. GitHub. https://github.com/CSSEGISandData/COVID-19.

[3] Hyndman, Rob J., and Yeasmin Khandakar. 2008. "Automatic Time Series Forecasting: The forecast Package for R." *Journal of Statistical Software* 27 (3): 1–22. https://doi.org/10.18637/jss.v027.i03.

[4] Hyndman, Rob J., George Athanasopoulos, Christoph Bergmeir, Gabriel Caceres, Leanne Chhay, Mitchell O'Hara-Wild, Fotios Petropoulos, Slava Razbash, Earo Wang, and Farah Yasmeen. 2026. *forecast: Forecasting Functions for Time Series and Linear Models*. R package version 9.0.2. https://doi.org/10.32614/CRAN.package.forecast.

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I acknowledge the use of Grammarly (https://www.grammarly.com) to check the spelling, grammar, and punctuation of the final text of this report.